

JEE MAIN REVISION MASTER: ELECTROMAGNETIC WAVES

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 166 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	EM Wave Characteristics & Spectrum	146 PYQs	EM Spectrum
Very High	Displacement Current & Maxwell's Equations	21 PYQs	Maxwell

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
EM Wave Characteristics & Spectrum	$c = \frac{1}{\sqrt{\mu_0 \epsilonpsilon_0}} = \frac{E_0}{B_0}, \quad u_{\text{avg}} = \epsilonpsilon_0 E_{\text{rms}}^2 = \frac{B_{\text{rms}}^2}{\mu_0}, \quad I = u_{\text{avg}} c = \frac{E_0 B_0}{2 \mu_0}$
Displacement Current & Maxwell's Equations	$I_d = \epsilonpsilon_0 \frac{d\Phi_E}{dt}, \quad \oint \vec{B} \cdot d\vec{l} = \mu_0 \left(I_c + \epsilonpsilon_0 \frac{d\Phi_E}{dt} \right)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 EM Wave Characteristics & Spectrum (Very High Yield — 146 PYQs)

Core Principle: Transverse nature of EM waves ($\vec{E} \perp \vec{B} \perp \vec{k}$), Poynting vector, energy density, and frequency bands of spectrum.

■■ **Caution / JEE Pitfall:** In an EM wave, the energy is distributed EQUALLY between electric and magnetic fields: $u_E = u_B$.

■ **Speed Trick:** Radiation pressure on perfect absorbing surface: $P = I/c$; on perfect reflecting surface: $P = 2I/c$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "An electromagnetic wave travels in free space along the x-direction. At a particular point in space and time, $B = 2 \times 10^{-7} \text{ T}$." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Match the LIST-I with LIST-II } List-I List-II A. Radio-wave I. is produced by Magnetron valve B. Micro-wave II. due to change in th..." [Ans: C]
- [2026 Morning Shift] Q#1 (MCQ): "The electric field in a plane electromagnetic wave is given by : $E(y) = 69 \sin[0.6 \times 10^3 x - 1.8 \times 10^{11} t] \text{ V/m}$. The expression fo..." [Ans: A]

3.2 Displacement Current & Maxwell's Equations (Very High Yield — 21 PYQs)

Core Principle: Maxwell's correction to Ampere's law, displacement current inside charging capacitor dielectric space.

■ **Caution / JEE Pitfall:** Inside the charging capacitor plates, conduction current is 0 and displacement current equals I_c ; in the wire, displacement current is 0.

■ **Speed Trick:** Continuity of current holds everywhere: $I_{\text{(conduction)}} = I_{\text{(displacement)}}$.

Representative Source PYQs:

- **[2026 Morning Shift] Q#1 (MCQ):** "A point light source emits E.M. waves in free space. A detector, placed at a distance of L_m , measures the intensity as I_o . The detector is now shifted to another location on the same spherical surface ensuring the angle between original location and new location as 45° . The measured intensity at new location will be ____." **[Ans: B]**
- **[2026 Evening Shift] Q#1 (Numerical):** "An electromagnetic wave of frequency 100 MHz propagates through a medium of conductivity, $\sigma = 10 \text{ mho/m}$. The ratio of maximum conduction current to displacement current is ____." **[Ans: 1800]**
- **[2026 Morning Shift] Q#3 (MCQ):** "A displacement current of 4.0 A can be set up in the space between two parallel plates of $6 \mu\text{F}$ capacitor. The rate of change of potential difference between the plates is ____." **[Ans: B]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

A point light source emits E.M. waves in free space. A detector, placed at a distance of L_m , measures the intensity as I_o . The detector is now shifted to another location on the same spherical surface ensuring the angle between original location and new location as 45° . The measured intensity at new location will be ____.

- (A) $\frac{I_o}{4}$
- (B) I_o **[Correct]**
- (C) $\frac{I_o}{\sqrt{2}}$
- (D) $\frac{I_o}{2}$

Correct Answer: B

Solution Analysis: The intensity (I) of electromagnetic waves from a point source in free space follows the inverse square law. $I = \frac{P}{4\pi r^2}$ Where P is the power of the source and r is the distance from the source. Initial distance is L and the intensity recorded is I_o . It is given that the detector is shifted to another location on the same spherical surface. On a spherical surface all points are at a constant distance r from the center (the source). Since the detector remains on the same spherical surface, its distance from the point source remains L . The angle (45°) represents

Benchmark Problem 2: JEE Main 2026 (Online) 24th January Morning Shift (Q#1)

Match the LIST-I with LIST-II } List-I List-II A. Radio-wave I. is produced by Magnetron valve B. Micro-wave II. due to change in the vibrational modes of atoms C. Infrared-wave III. due to inner shell electrons moving from higher energy level to lower energy level D. X-ray IV. due to rapid acceleration of electrons Choose the correct answer from the options given below:

- (A) -IV, B-II, C-I, D-III
- (B) -IV, B-III, C-I, D-II
- (C) -IV, B-I, C-II, D-III **[Correct]**
- (D) -II, B-IV, C-III, D-I

Correct Answer: C

Solution Analysis: Radio-waves 1. Radio waves have the longest wavelengths in the EM spectrum. They are artificially generated by transmitters. The core process involves an oscillating electric current in a special conductor called an antenna. 2. According to electromagnetic theory, an accelerating electric charge produces electromagnetic waves. In an antenna, electrons are rapidly accelerated back and forth. 3. So, the radio waves are produced due to rapid acceleration of electrons. Micro-waves 1. Micro-waves are high-frequency radio waves. While they can be produced by acceleration, high-power microwaves (like